PT MERDEKA TSINGSHAN INDONESIA

STANDARD OPERATING PROCEDURE
SPONGE COPPER REPLACEMENT POST
海绵铜置换岗位

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1. Purpose 目的

This procedure covers the operator's responsibility for visual inspection, start up and shutdown of the sponge copper replacement post under complete closure.

此程序涵盖操作员对完全关闭下的海绵铜置换岗位进行目视操作前检查、启动运行和停车的责任

2. Functional Description 功能说明

It is the responsibility of the operator to ensure that the sponge copper replacement post is visually inspected before and after each shift to determine whether it can start and operate safely, or whether other activities (such as maintenance and repair or other operator tasks) must be performed before starting, to determine the sequential operation method of the sponge copper replacement post and smooth operation, Stop order of normal stops and emergency stops.

操作员有责任确保在从停机启动之前和每个班次开始时对海绵铜置换岗位进行目视检查，确定是否可以启动并安全运行，或者是否必须在开始启动之前执行其他活动(例如维护修理或其他操作员任务)，确定海绵铜置换岗位的按照顺序启动的操作方法及平稳运行，正常停车和紧急停车的停止顺序。

3. References 参考

- MTI-CP-SOP-Post Technical Operation Procedures.
  MTI-CP-SOP岗位操作规程
- Neutralizing Tank (4402-TNK-002/003).
  中和槽 (4402-TNK-002/003)
- Neutralizing Pump (4402-PSL-001/002).
  中和泵 (4402-PSL-001/002)
- Gypsum Filter Press (4402-FIL-007/008/009).
  石膏压滤机 (4402-FIL-007/008/009)
- Washing Water Tank (4402-TNK-017).
  洗涤水槽 (4402-TNK-017)
- Washing Water Pump (4402-PSO-014).
  洗涤水泵 (4402-PSO-014)
- Solution Tank After Gypsum Centrifugal Separation (4402-TNK-004).
  石膏离心分离后液槽 (4402-TNK-004)
- Solution Pump After Gypsum Centrifugal Separation (4402-PSO-003/004).
  石膏离心分离后液泵 (4402-PSO-003/004)
- Iron Power Replacement Tank (4402-TNK-005/006).
  铁粉置换槽 (4402-TNK-005/006)
- Iron Power Replacement Pump (4402-PSL-003/004).
  铁粉置换泵 (4402-PSL-003/004)
- Sponge Copper Filter Press (4402-FIL-001/002).

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海绵铜压滤机 (4402-FIL-001/002)

- Solution Tank After Iron Power Replacement (4402-TNK-007).
  铁粉置换后液槽 (4402-TNK-007)
- Solution Pump After Iron Power Replacement (4402-PSO-005/006).
  铁粉置换后液泵 (4402-PSO-005/006)
- Belt Conveyor (4402-BCV-002/003/004).
  皮带机 (4402-BCV-002/003/004)
- Screwed Conveyor (4402-CNV-001/002/003).
  螺旋输送机 (4402-CNV-001/002/003)
- Sponge Copper Ton Bag Packer (4402-BPR-001/002/003).
  海绵铜吨袋包装机 (4402-BPR-001/002/003)

4. PPE 个人保护装备



5. Hazards & Control 危害与控制

Description of Hazard / Environmental Issue 危害	Control Strategy 控制
1. Mechanical injury; 2. Electric shock; 3. Slippery conditions and a trip to the danger; 4. Liquid corrosion; 5. Object falling from high altitudes; 6. Vehicle collision crush; 7. Explode; 8. Liquid sputtering; 9. Noise. 1. 机械伤害; 2. 触电; 3. 湿滑的条件，绊倒危险;	1. Install a protective cover on the rotating part of the equipment; 2. Do not contact the electrical switch with wet hands during operation, Regular inspection of the equipment leakage protection device is complete, effective; 3. Please hold the handrail up and down the stairs; 4. Wear appropriate labor protection supplies, once the skin contact with hydrochloric acid solution, immediately rinse with a lot of water; 5. Wear labor protection goods and check the safety of stacking items on the operation platform; 6. Walk along the prescribed route, and stay away from the sponge copper transport

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Description of Hazard / Environmental Issue 危害	Control Strategy 控制
4. 液体腐蚀； 5. 高空坠物； 6. 车辆碰撞碾压； 7. 爆炸； 8. 液体喷溅； 9. 噪音。	vehicles; 7. No open fire, no sparks, no smoking, and strengthen ventilation; 8. Wear protective masks and gloves, as well as other protective products as required； 9. Wear the earplugs to protect your hearing. 1. 设备的转动部位安装防护罩； 2. 操作时不要湿手接触电器开关，定期检查设备漏电保护装置齐全、有效； 3. 上下楼梯请扶好扶手； 4. 一旦皮肤接触到盐酸溶液，立即用大量清水冲洗； 5. 穿戴好劳保护品，检查各平台的物品堆放安全性； 6. 按照规定的路线行走，远离海绵铜运输车辆； 7. 禁止明火、禁止火花、禁止吸烟，加强通风； 8. 按要求戴好防护面具和防护手套，以及其它劳保护品； 9. 佩戴好耳塞，保护听力。

6. Equipment 工具/ 设备

4402-AGT-001/002	Neutralizing Tank Agitator 中和槽搅拌器
4402-BIN-001	Iron Power Bin 铁粉仓
4402-ADG-001/002	Rotary Weighing Feeding Valve 旋转称重给料阀
4402-AGT-003/004	Iron Power Replacement Tank Agitator 铁粉置换槽搅拌器
	Clean the site tools 清扫现场工具

7. Procedure 程序

№	Stage	Key Points
1	General Check 普查	1. Confirm the pre-job technical training; 2. Wear suitable labor protection products, check and confirm that there is no unsafe state or safety risks on the site;

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№	Stage	Key Points
		3. Check and clean the equipment and the site sanitation; 4. Contact with the process before and after, and give clear instructions. 1. 确认经过岗前技术培训; 2. 穿戴好适合的劳保护品，检查确认现场没有存在不安全状态或安全隐患; 3. 检查并打扫设备、现场卫生; 4. 与前后工序取得联系，并发出明确指令。
2	Inspect Solution Tank After Gypsum Centrifugal Separation (4402-TNK-004), Iron Power Replacement Tank (4402-TNK-005/006), Solution Tank After Iron Power Replacement (4402-TNK-007), Neutralizing Tank (4402-TNK-002/003), Washing Water Tank(4402-TNK-017) 检查石膏离心分离后液槽(4402-TNK-004)，铁粉置换槽(4402-TNK-005/006)，铁粉置换后液槽(4402-TNK-007)，中和槽(4402-TNK-002/003)，洗涤水槽(4402-TNK-017)	1. Check whether the inlet and outlet pipelines and auxiliary pipelines, whether the installation of flange and valve meet the requirements, and whether the slurry storage tank leaks liquid; 2. Check the level control assembly is normal. 1. 检查出入口管线及附属管线，法兰、阀门安装是否符合要求，储浆槽是否漏液; 2. 检查液位控制组件是否正常。
3	Inspect Solution Pump After Gypsum Centrifugal Separation(4402-PSO-003/004), Iron Power Replacement Pump(4402-PSL-003/004), Solution Pump After Iron Power Replacement(4402-	1. Check whether the pump inlet pipeline and auxiliary pipeline, flange and valve installation meet the requirements, whether the anchor bolts and ground wire are good, and whether the coupling is installed; 2. Turn the pump shaft 3 times, should be flexible without block; 3. Check whether the lubricating oil level is normal. The oil level should be between 1/2 and 2/3 of the oil lens; 4. Check the sealing condition of the pump is normal; 5. Point to move to confirm the correct steering; 6. Check whether the voltage meter, signal lamp and other

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№	Stage	Key Points
	PSO-005/006), Neutralizing Pump(4402-PSL-001/002), Washing Water Pump(4402-PSO-014) 检查石膏离心分离后液泵(4402-PSO-003/004)，铁粉置换泵(4402-PSL-003/004)，铁粉置换后液泵(4402-PSO-005/006)，中和泵(4402-PSL-001/002)，洗涤水泵(4402-PSO-014)	meters of the pump control cabinet are normal; 7. Check whether the pump unit has air, if so, should be excluded; 8. Check that the pump head is under the liquid levels of the various tanks. 1. 检查泵出入口管线及附属管线，法兰、阀门安装是否符合要求，地脚螺栓及地线是否良好，联轴器及防护罩是否装好; 2. 手转动泵轴3圈，应灵活无阻滞; 3. 检查润滑油油位是否正常，油位应在油镜的1/2 ~ 2/3之间; 4. 检查泵的密封状况是否正常; 5. 点动确认转向正确; 6. 检查泵控制柜电压表、信号灯等仪表指示是否正常; 7. 检查泵机组是否有空气，如有应予以排除; 8. 检查泵头是否处于各种槽的液位下。
4	Inspect Sponge Copper Filter Press(4402-FIL-001/002), Gypsum Filter Press(4402-FIL-007/008/009) 检查海绵铜压滤机(4402-FIL-001/002)，石膏压滤机(4402-FIL-007/008/009)	1. Check whether the number of filter plates is sufficient, whether there is damage, whether the filter plate is clean, and whether the placement meets the requirements; 2. Check whether the filter cloth is folded, whether it is damaged, and whether the filter performance is good; 3. Check whether the pressure gauge of the mine pipeline is normal; 4. Check whether the connection is close and whether there is leakage; 5. Check whether the oil of the filter press is sufficient and whether the oil level meets the requirements (1 / 2~1 / 3); 6. Check whether the hydraulic oil top device is normal; 7. Check whether the emergency stop device is normal; 8. Check whether the pressure gauge of the hydraulic oil is normal and whether the pointer is in the specified interval; 9. Check whether the import and export of the filter press are smooth; 10. Automatic plate drawing device; 11. Whether the water connection plate is flexible and easy to use. 1. 检查滤板数量是否足够，有无破损，滤板是否清洁，安放是否符合要求; 2. 检查滤布是否折叠，有无破损，过滤性能是否良好;

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		3. 检查给矿管路压力表是否正常; 4. 检查各处联接是否紧密，有无泄露; 5. 检查压滤机油是否足够，油位是否符合要求(1/2 ~ 1/3); 6. 检查液压油顶装置是否正常; 7. 检查急停装置是否正常; 8. 检查液压油的压力表是否正常，指针是否在规定的区间; 9. 检查压滤机各进出口是否畅通; 10. 检查自动拉板装置良好; 11. 接水盘是否灵活好用。
5	Inspect Belt Conveyor(4402-BCV-002/003/004) 检查皮带机(4402-BCV-002/003/004)	1. Each bearing is well lubrication and closed, the reducer oil level is appropriate, the upper and lower roller lubrication is good, the operation is flexible, whether the wear of the first and tail wheels and the tape are filled with materials, and whether the cleaning device of the first and tail wheels is in good condition; 2. Comprehensively check whether all parts of the belt conveyors are intact, whether the tension of the tape is appropriate, and whether the corresponding tape tightening device and dust-proof facilities are in good condition; 3. Whether the electrical facilities and control system are intact, whether the protective device is firm, and whether the funnel baffle and liner are damaged; 4. Check whether the cushion rollers of the impact load parts are intact; 5. Check the installation of belt sweeper; 6. Whether the installed electrical facilities and signal devices are intact and responsive; 7. Place the field control box in the manual position, press the start button, slowly rotate the potentiometer, and watch whether the running direction of the tape is correct, and whether the belt can run away; 8. After checking and contacting the post, press the start button; 9. During the operation process, check whether the tape machine emergency stop device, belt deviation correction and speed measuring sensor are in good condition. 1. 各部轴承润滑与封闭良好，减速机油位适当，上下托滚润滑良好，运转灵活，首、尾轮磨损情况与胶带间

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№	Stage	Key Points
		是否夹有物料，首、尾轮的清扫装置是否完好； 2. 全面检查皮带机各部分是否完好无损，胶带的张紧度是否合适，相应的胶带拉紧装置及防尘设施是否完好； 3. 电器设施和控制系统是否完好，防护装置是否牢固，漏斗挡板和衬板是否损坏； 4. 检查承受冲击载荷部位的缓冲托辊是否完好； 5. 检查皮带机清扫器安装是否正常； 6. 装设的电器设施及信号装置是否完好和反应灵敏； 7. 将现场控制盒置于手动位置，按一下启动按钮，迟缓旋动电位器，观看胶带运转方向能否正确，皮带是否跑偏； 8. 检查无误并与前后岗位联系后，按下启动按钮； 9. 运转过程中，检查皮带机急停装置、皮带纠偏和测速传感器是否控制灵敏。
6	Inspect Screwed Conveyor(4402-CNV-001/002/003) 检查螺旋输送机(4402-CNV-001/002/003)	1. Check whether the lubricating grease in the bearing is appropriate; 2. Check whether the oil quality and quantity of the lubricating oil in the reducer are normal; 3. Check whether the bolts are fastened; 4. Check whether the cover plate of the screw conveyor must be tightly covered and whether the conveyor is clean; 5. Check whether the spiral blade is deformed ; 6. Determine whether the rotation direction of the motor meets the work requirements. 1. 检查轴承内润滑脂是否合适； 2. 检查减速机内的润滑油油质、油量是否正常； 3. 检查各螺栓是否紧固； 4. 检查螺旋输送机的盖板必须盖严，输送机内是否清理干净； 5. 检查螺旋叶片是否变形； 6. 判明电机旋转方向是否符合工作要求。
7	Inspect Sponge Copper Ton Bag Packer(4402-BPR-001/002/003) 检查海绵铜吨袋包装机(4402-BPR-001/002/003)	a. Check whether the safety shield of each transmission part is intact and effective; b. Check whether the lubrication part has been grease; c. Check whether the indicator instrument is working normally, and the signal is sensitive and effective; d. Check whether the material scraping and fixing screw is loose; e. Check whether the pneumatic butterfly valve is normal

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		and flexible; f. Check whether the lifting scale body platform is normal; g. Check whether the automatic uncoupling device is normal; h. Check whether the pneumatic clip is normal; i. Check the control system for normal operation. a. 检查各传动部位安全防护罩是否完好有效; b. 检查需要润滑部位是否已加油脂; c. 检查指示仪表工作是否正常，信号灵敏有效; d. 检查料刮固定螺丝是否松动; e. 检查气动阀是否正常、灵活; f. 检查升降秤体平台是否正常; g. 检查自动脱钩装置是否正常; h. 检查气动夹袋器是否正常; i. 检查控制系统是否正常。
8	Check the other Equipment and Facilities 检查其它设备设施	1. Neutralizing tank agitator(4402-AGT-001/002), Iron Power Replacement Tank Agitator(4402-AGT-003/004) a. Check whether the screws of the mixing paddle are fastened; b. Check whether the positive and reverse version of the agitator is correct; c. Check the surface of the mixer is smooth, without rough and pitting phenomenon; d. Check whether the mechanical seal of the mixer is good and whether the oil seal is normal; e. Electric mixer idling, check for abnormal sound, whether the up and down are concentric or shaking. 1. 中和槽搅拌器(4402-AGT-001/002)，铁粉置换槽搅拌器(4402-AGT-003/004) a. 检查搅拌桨的螺丝是否紧固; b. 检查搅拌器的正反转是否正确; c. 检查搅拌器的搅拌桨表面光滑，无粗糙和点蚀现象; d. 检查搅拌器的机械密封是否良好，各油封是否正常; e. 电动搅拌器进行空转，检查是否有异常声响，上下是否同心或摇晃。 2. Iron Power Bin(4402-BIN-001) a. Check whether the inlet and outlet pipes and auxiliary

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		pipes and flanges of each warehouse meet the requirements, and whether the seal of each warehouse is in good condition; b. Check the material level control assembly for normal operation. 2. 铁粉仓(4402-BIN-001) a. 检查各仓出入口管路及附属管路、法兰是否符合要求，各仓密封是否完好； b. 检查料位控制组件是否正常。 3. Chute a. Check sponge copper filter press(4402-FIL-001/002), Gypsum filter press(4402-FIL-007/008/009) whether the feeding chute is in good condition; b. Check sponge copper filter press(4402-FIL-001/002), gypsum filter press(4402-FIL-007/008/009) whether the blocking detection switch of the feeding chute is normal. 3. 溜槽 a. 检查海绵铜压滤机(4402-FIL-001/002)、石膏压滤机(4402-FIL-007/008/009)下料溜槽是否完好； b. 检查海绵铜压滤机(4402-FIL-001/002)、石膏压滤机(4402-FIL-007/008/009)下料溜槽的堵料检测开关是否正常。 4. Rotary Weighing Feeding Valve(4402-ADG-001/002) a. Check whether the lubricating oil of the motor reducer is normal, and whether the motor rotation direction is correct; b. Check whether the control cabinet system is normal; c. Check whether the weighing sensor and speed measurement sensor are normal and sensitive; d. Check whether the feeding inlet and outlet of the feeding valve are blocked. 4. 旋转称重给料阀(4402-ADG-001/002) a. 检查电机减速机润滑油是否正常，电机旋转方向是否正确； b. 检查控制柜系统是否正常； c. 检查称重传感器、测速传感器是否正常、灵敏； d. 检查给料阀进料口、出料口是否堵塞。

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		5. Valve a. Check the washing sink (4402-TNK-017) new water a gate valve (50GA), the bottom discharge of four butterfly valve (100BF,100BF,100BF,100BF) is normal and flexible; b. Check whether two butterfly valve (125BF,125BF) at the inlet of the washing water pump (4402-PSO-014) and two butterfly valve (100BF,100BF) is normal and flexible; c. Check the two butterfly valves (100BF, 100BF) from the lime milk delivery pump (4524-PSL-003/004) (4402-TNK-002/003), Calcium chloride solution pump (4402-PSO-001/002) pumps two butterfly valves (200BF, 200BF), The bottom drain has two butterfly valves (100BF, 100BF), Whether the two butterfly valves (125BF, 125BF) pumped by the neutralization pump (4402-PSL-001/002) are normal and flexible; d. Check whether the four butterfly valves (200BF, 200BF, 200BF) (4402-PSL-001/002) and the two butterfly valves (125BF, 125BF) are normal and flexible; e. Check Washing water pump (4402-PSO-014) (100BF, 100BF, 100BF, 100BF, 100BF, 100BF) of gypsum filter press (4402-FIL-007/008/009), Three butterfly valves (125BF, 125BF, 125BF) pumped by the neutralization pump (4402-PSL-001/002), Six ball valves with compressed air (4601-CAT-001/002/003/004) (50BA, 50BA, 50BA, 100BA, 100BA, 100BA), Of the three butterfly valves (100BF, 100BF, 100BF), Whether the three butterfly valves (100BF, 100BF, 100BF) after washing liquid are normal and flexible; f. Check three butterfly valve from the gypsum filter press (4402-FIL-007/008/009) (250BF,250BF,250BF) of after gypsum centrifugation separation tank (4402-TNK-004),Whether three butterfly valve (100BF,100BF,100BF) is normal and flexible; g. Check the three butterfly valves (4402-PSO-003/004) (300BF, 300BF, 300BF), Whether the two butterfly valves (250BF, 250BF) are normal and flexible; h. Check the two butterfly valves (250BF, 250BF) of the gypsum centrifugal separation liquid pump (4402-PSO-003/004) from the iron powder replacement tank (4402-TNK-005/006), Whether the two butterfly valves (100BF, 100BF) at the bottom are intact and flexible; i. Check the inlet of four butterfly valves (4402-PSL-

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		003/004) (200BF, 200BF, 200BF, 200BF), Whether the two butterfly valves (125BF, 125BF) are normal and flexible; j. Check the four butterfly valves (100BF, 100BF, 100BF, 100BF) of the sponge copper filter (4402-FIL-001/002) gold mud post (4402-PSO-015), Two butterfly valves (125BF, 125BF) pumped by the iron powder replacement pump (4402-PSL-003/004), four ball valves with compressed air (4601-CAT-001/002/003/004) (50BA, 50BA, 100BA, 100BA), Two butterfly valves (100BF, 100BF), Whether the two butterfly valves (100BF, 100BF) after washing liquid are normal and flexible; k. Check one valve (250SP) from the sponge copper filter (4402-FIL-001/002) (4402-TNK-007) after iron powder replacement, Whether the two butterfly valves (100BF, 100BF) at the bottom are intact and flexible; l. Check whether the three butterfly valves (300BF, 300BF, 300BF) at the inlet of the liquid pump (4402-PSL-005/006) and the two butterfly valves (250BF, 250BF) are normal and flexible. 5. 阀门 a. 检查洗涤水槽(4402-TNK-017)新水一个闸阀(50GA)，底部排液一个蝶阀(100BF)是否正常、灵活; b. 检查洗涤水泵(4402-PSO-014)入口三个蝶阀(125BF,125BF,125BF)，出口两个蝶阀(100BF,100BF)是否正常、灵活; c. 检查中和槽(4402-TNK-002/003)的来自石灰乳输送泵(4524-PSL-003/004)泵过来石灰乳的两个蝶阀(100BF,100BF)，氯化钙溶液泵(4402-PSO-001/002)泵过来氯化钙溶液的两个蝶阀(200BF,200BF)，底部排液两个蝶阀(100BF,100BF)，中和泵(4402-PSL-001/002)泵过来的两个蝶阀(125BF,125BF)是否正常、灵活; d. 检查中和泵(4402-PSL-001/002)入口四个蝶阀(200BF,200BF,200BF,200BF)，出口两个蝶阀(125BF,125BF)是否正常、灵活; e. 检查石膏压滤机(4402-FIL-007/008/009)的洗涤水

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		泵(4402-PSO-014)泵过来的洗涤水六个蝶阀(100BF,100BF,100BF,100BF,100BF,100BF)，中和泵(4402-PSL-001/002)泵过来的三个蝶阀(125BF,125BF ,125BF)，压缩空气(4601-CAT-001/002/003/004)的六个球阀(50BA,50BA,50BA,100BA,100BA,100BA)，滤液三个蝶阀(100BF,100BF,100BF)，洗涤后液三个蝶阀(100BF,100BF,100BF)是否正常、灵活； f. 检查石膏离心分离后液槽(4402-TNK-004)的石膏压滤机(4402-FIL-007/008/009)过来的滤液三个蝶阀(250BF,250BF,250BF)，底部排液三个蝶阀(100BF,100BF,100BF)是否正常、灵活； g. 检查石膏离心分离后液泵(4402-PSO-003/004)入口三个蝶阀(300BF,300BF,300BF)，出口两个蝶阀(250BF,250BF)是否正常、灵活； h. 检查铁粉置换槽(4402-TNK-005/006)的石膏离心分离后液泵(4402-PSO-003/004)泵过来的两个蝶阀(250BF,250BF)，底部排液两个蝶阀(100BF,100BF)是否完好、灵活； i. 检查铁粉置换泵(4402-PSL-003/004)入口四个蝶阀(200BF,200BF,200BF,200BF)，出口两个蝶阀(125BF,125BF)是否正常、灵活； j. 检查海绵铜压滤机(4402-FIL-001/002)金泥岗位洗涤水泵(4402-PSO-015)泵过来的洗涤水蝶阀四个蝶阀(100BF,100BF,100BF,100BF)，铁粉置换泵(4402-PSL-003/004)泵过来的两个蝶阀(125BF,125BF)，压缩空气(4601-CAT-001/002/003/004)的四个球阀(50BA,50BA,100BA,100BA)，滤液两个蝶阀(100BF,100BF)，洗涤后液两个蝶阀(100BF,100BF)是否正常、灵活； k. 检查铁粉置换后液槽(4402-TNK-007)的海绵铜压滤机(4402-FIL-001/002)过来的滤液一个阀门(250SP)，底部排液两个蝶阀(100BF,100BF)是否完

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		好、灵活； l. 检查铁粉置换后液泵(4402-PSO-005/006)入口三个蝶阀(300BF,300BF,300BF)，出口两个蝶阀(250BF,250BF)是否正常、灵活。
9	Inspect Materials Level 检查物料料位	1. Check whether the iron powder bin (4402-BIN-001) is sufficient; 2. Check whether the liquid level of the solution tank after gypsum centrifugal separation (4402-TNK-004), iron power replacement tank (4402-TNK-005/006), solution tank after iron power replacement (4402-TNK-007), chlorinated solution tank (4402-TNK-001), neutralizing tank (4402-TNK-002/003) and washing water tank(4402-TNK-017)meets the pressure requirements; 3. Check whether the material is stored in the sponge copper filter press(4402-FIL-001/002) and gypsum filter press(4402-FIL-007/008/009), whether the hydraulic oil is sufficient, and whether there is a filter cake in the feeding chute； 4. Check whether the material is stored on the tape machine(4402-BCV-002/003/004), and there is no adhesive deposit on the surface of the head and tail wheel and the belt joint； 5. Check whether the material is stored in the screw conveyor(4402-CNV-001/002/003). 1. 检查铁粉仓(4402-BIN-001)存料是否充足； 2. 检查石膏离心分离后液槽(4402-TNK-004)、铁粉置换槽(4402-TNK-005/006)、铁粉置换后液槽(4402-TNK-007)、氯化溶液储槽(4402-TNK-001)，中和槽(4402-TNK-002/003)和洗涤水槽(4402-TNK-017)液位是否达到要求； 3. 检查海绵铜压滤机(4402-FIL-001/00)、石膏压滤机(4402-FIL-007/008/009)内是否存料，液压油是否足够，下料溜槽是否存有滤饼； 4. 检查皮带机(4402-BCV-005)上是否存料，首尾轮表面与皮带结合部无黏附积矿存在； 5. 检查螺旋输送机(4402-CNV-001/002/003)内是否存料。
10	Start-Up Of the Sponge Copper	1. No-load start should be guaranteed when starting; 2. The driving sequence is along the material direction, from

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	Replacement Post 海绵铜置换岗位开机	the back to drive that is, the reverse driving, taking into account to reduce the idling of the equipment, each equipment has been delayed to open; At the same time, Confirm that the process behind the sponge copper replacement post has been started and has the feeding conditions: Open the washing sink (4402-TNK-017) new water a gate valve (50GA), and close the new water⟹ Open the desulfurization tower (4306-TOW-004) (65BA), open the two butterfly valves (200BF, 200BF) pumping over the chloride solution pump (4402-PSO-001/002), to neutralize tank (4402-TNK-002/003) filled with desulfurization tower (4306-TOW-004) or after the solution of the storage tank (4402-TNK-001) pump (4402-PSO-001/002), start the neutralizing tank agitator (4402-AGT-001/002)⟹ Open the two butterfly valves (100BF, 100BF) coming from the lime milk preparation post, Inform the lime milk preparation post, start the lime milk conveying pump (4524-PSL-003/004), and transport the lime milk to the neutralization tank (4402-TNK-002/003)⟹Start the gypsum filter press (4402-FIL-007/008/009)After neutralization, press the gypsum filter press (4402-FIL-007/008/009), The oil pump starts to work, and then press the "press" button, the piston pushes the pressing plate pressure, when the pressure force reaches the high pressure, the hydraulic system automatically stop and maintain the pressure, Open the three butterfly valves (125BF, 125BF, 125BF) pumped from the neutralization pump (4402-PSL-001/002), Of the three butterfly valves (100BF, 100BF, 100BF), After three butterfly valves (100BF, 100BF, 100BF)⟹ Open the two butterfly valves at the outlet (125BF, 125BF), Start the neutralizing pump (4402-PSL-001/002), Open the inlet of four butterfly valves (4402-PSL-001/002) (200BF, 200BF, 200BF, 200BF)⟹ The slurry is admitted into each filter chamber through the feed hole on the thrust plate, and is filtered under the feed pressure not greater than 0.6MPa to gradually form the filter cake⟹ After the filter press filter cake is filled, Close the valves (4402-PSL-001/002) (200BF, 200BF, 200BF, 200BF), Stop the neutralizing pump (4402-PSL-001/002) and close the outlet two butterfly valves (125BF, 125BF)⟹ Open the washing water pump (4402-PSO-014) with two outlet butterfly valves (100BF, 100BF), Start the washing water pump (4402-PSO-014), Open the washing water pump (4402-PSO-014) into an outlet butterfly valve (125BF,

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		125BF), Wash the filter cake at a pressure not greater than 0.6MPa. After the operation, Close the washing water pump (4402-PSO-014) into the outlet butterfly valve (125BF, 125BF), Stop the washing water pump (4402-PSO-014), Close the washing water pump (4402-PSO-014) with two outlet butterfly valves (100BF, 100BF), Close the gypsum filter press (4402-FIL-007/ 008/009) wash six butterfly valves (100BF, 100BF, 100BF, 100BF, 100BF, 100BF)⟹ Six ball valves (50BA, 50BA, 50BA, 100BA, 100BA, 100BA) slowly opening compressed air (4601-CAT-001/002/003/004), Dry the filter cake at a pressure not greater than 0.6MPa. After operation, close the compressed air six ball valves (50BA, 50BA, 50BA, 100BA, 100BA, 100BA)⟹ Press and hold the "release" button on the operation panel, and the pressure plate will automatically stop after falling back in place, and the pull plate hook moves forward to unload the reciprocating pull plate⟹The filter press (4402-FIL-007/008/009) flows to the washing tank (4402-TNK-017), After the washing sink (4402-TNK-017) is too fully filled, open the washing water pump (4402-PSO-014) and exit two butterfly valve (100BF,100BF), Start the washing pump (4402-PSO-014), open three butterfly valve (125BF,125BF,125BF) at the inlet of the washing pump (4402-PSO-014), pump the washer recycling tank (4306-TNK-007)⟹Open three butterfly valve from the gypsum filter press (4402-FIL-007/008/009) (250BF,250BF,250BF) of after gypsum centrifugation separation tank (4402-TNK-004),Whether three butterfly valve (100BF,100BF,100BF), The filtrate flows to the gypsum centrifuge after separation tank (4402-TNK-004), After the liquid tank (4402-TNK-004) is filled, Add iron powder (4402-BIN-001) to sufficient iron powder⟹ Start the iron powder replacement tank mixer (4402-AGT-003/004), Open the two butterfly valves (250BF, 250BF) of the liquid pump (4402-PSO-003/004), Start the liquid pump (4402-PSO-003/004), Open the inlet of the liquid pump (4402-PSO-003/004) (300BF, 300BF, 300BF), Pump the filtrate into the iron powder replacement tank (4402-TNK-005/006)⟹ After the iron powder replacement tank (4402-TNK-005/006) is filled, Open the rotating weighing feeding valve (4402-ADG-001/002), slowly add iron powder to the iron powder replacement tank (4402-TNK-005/006)⟹Start the sponge copper filter press (4402-FIL-001/002)After the replacement, press the "Start" button (4402-FIL-001/002), The oil pump starts to

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		work, and then press the "press" button, the piston pushes the pressing plate pressure, when the pressure force reaches the high pressure, the hydraulic system automatically stop and maintain the pressure, Open the four butterfly valves (100BF, 100BF, 100BF, 100BF) pumped by the gold clay washing water pump (4402-PSO-015), Two butterfly valves (125BF, 125BF) pumped from the iron powder replacement pump (4402-PSL-003/004), Two butterfly valves (100BF, 100BF) and two butterfly valves (100BF, 100BF)⟹ Open the two butterfly valves (125BF,125BF)) (4402-PSL-003/004), Start the iron powder replacement pump (4402-PSL-003/004) and open the inlet four butterfly valves (200BF, 200BF, 200BF, 200BF)⟹ The slurry is admitted into each filter chamber through the feed hole on the thrust plate, and is filtered under the feed pressure not greater than 0.6MPa to gradually form the filter cake⟹ After the filter press filter cake is filled, Close the pump (4402-PSL-003/004) inlet four butterfly valves (200BF, 200BF, 200BF, 200BF), Stop the iron powder replacement pump (4402-PSL-003/004), Close two butterfly valves (125BF,125BF) (4402-PSL-003/004)⟹Open the two butterfly valves (100BF, 65BF) of the washing pump (4402-PSO-015), Start the washing water pump (4402-PSO-015), Open the two butterfly valves (125BF, 125BF) at the inlet of the washing water pump (4402-PSO-015), Wash the filter cake at a pressure not greater than 0.6MPa. After the operation, Close the two butterfly valves (125BF, 125BF) at the inlet of the washing water pump (4402-PSO-015), Stop the gold clay washing water pump (4402-PSO-015), Close the two butterfly valves (100BF, 65BF) at the outlet of the washing water pump (4402-PSO-015), Close the sponge copper press filter (4402-FIL-001/002) wash water four ball valves (100BF, 100BF, 100BF, 100BF)⟹ Four ball valves (50BA, 50BA, 100BA,100BA) on compressed air (4601-CAT-001/002/003/004), Dry the filter cake at a pressure not greater than 0.6MPa. After the operation, close the four compressed air ball valves (50BF, 50BF, 100BF, 100BF)⟹ Press and hold the "release" button on the operation panel, the pressure plate will automatically stop after falling back in place, and the pull plate hook goes forward to unload the reciprocating pull plate⟹ Sponge copper press filter (4402-FIL-001/002) filtrate flow to the liquid tank after replacement of iron powder (4402-TNK-007), After the tank (4402-TNK-007) is filled with iron powder, Open the two butterfly valves

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		(250BF, 250BF) of the outlet of the liquid pump (4402-PSO-005/006), Start the liquid pump after iron powder replacement (4402-PSO-005/006), Open three butterfly valves (300BF, 300BF, 300BF) (4402-PSO-005/006), Pump the filtrate to the PP area for the iron removal process ⟹ Sponge copper press filter (4402-FIL-001/002) washing water flow to the gold mud post washing tank (4402-TNK-018) ⟹ During the filter cake discharge process, start the spiral conveyor (4402-CNV-001/002/003) ⟹ Start the sponge copper ton bag packaging machine (4402-BPR-001 / 002 / 003) ⟹ After the packing machine, start the belt machine (4402-BCV-002 / 003 / 004), and the forklift fork takes the ton bag. 1. 启动时应保证空载启动； 2. 开车顺序为沿物料走向，从后往前开即逆向开车，兼顾减少设备的空转，各设备相继延时开启；同时，确认海绵铜置换岗位后面的工序已经开车并具备给料条件： 打开洗涤水槽(4402-TNK-017)新水一个闸阀(50GA)，并注满新水后关闭⟹打开脱硫塔(4306-TOW-004)外排石膏溶液蝶阀一个(65BA)，打开烟气洗涤岗位氯化溶液泵(4402-PSO-001/002)泵过来氯化钙溶液的两个蝶阀(200BF,200BF)，待中和槽(4402-TNK-002/003)充满尾气吸收岗位脱硫塔(4306-TOW-004)的石膏溶液或烟气洗涤岗位氯化溶液储槽(4402-TNK-001)泵(4402-PSO-001/002)过来的溶液后，启动中和槽搅拌器(4402-AGT-001/002)⟹打开石灰乳制备岗位过来的石灰乳两个蝶阀(100BF,100BF)，通知石灰乳制备岗位，启动石灰乳输送泵(4524-PSL-003/004)，输送石灰乳至中和槽(4402-TNK-002/003)⟹启动石膏压滤机(4402-FIL-007/008/009)中和结束后，按下石膏压滤机(4402-FIL-007/008/009)“启动”按钮，油泵开始工作，然后再按一下“压紧”按钮，活塞推动压紧板压紧，当压紧力到达调定高点压力后，液压系统自动跳停并保持压力，打开中和泵(4402-PSL-001/002)泵过来的三个蝶阀(125BF,125BF ,125BF)、滤液三个蝶阀

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		(100BF,100BF,100BF)、洗涤后液三个蝶阀(100BF,100BF,100BF)⟶打开中和泵(4402-PSL-001/002)出口两个蝶阀(125BF,125BF)，启动中和泵(4402-PSL-001/002)，打开中和泵(4402-PSL-001/002)入口四个蝶阀(200BF,200BF,200BF,200BF)⟶料浆通过止推板上的进料孔进去各个滤室，在进料压力不大于0.6MPa下加压过滤，逐渐形成滤饼⟶压滤机滤饼充满后，关闭中和泵(4402-PSL-001/002)入口四个蝶阀(200BF,200BF,200BF,200BF)，停止中和泵(4402-PSL-001/002)，关闭中和泵(4402-PSL-001/002)出口两个蝶阀(125BF,125BF)⟶打开洗涤水泵(4402-PSO-014)两个出口蝶阀(100BF,100BF)，启动洗涤水泵(4402-PSO-014)，打开洗涤水泵(4402-PSO-014)入个出口蝶阀(125BF,125BF)，在压力不大于0.6MPa下对滤饼进行洗涤，操作完成后，关闭洗涤水泵(4402-PSO-014)入个出口蝶阀(125BF,125BF)，停止洗涤水泵(4402-PSO-014)，关闭洗涤水泵(4402-PSO-014)两个出口蝶阀(100BF,100BF)，关闭石膏压滤机(4402-FIL-007/008/009)洗涤水六个蝶阀(100BF,100BF,100BF,100BF,100BF,100BF)⟶缓慢打开压缩空气(4601-CAT-001/002/003/004)的六个球阀(50BA,50BA,50BA,100BA,100BA,100BA)，在压力不大于0.6MPa下对滤饼进行风干，操作完成后，关闭压缩空气六个球阀(50BA,50BA,50BA,100BA,100BA,100BA)⟶按住操作面板上的“松开”按钮，压板退回到位后会自动停止，同时拉板钩前进进行往复拉板卸料⟶石膏压滤机(4402-FIL-007/008/009)滤饼洗涤液流向洗涤水槽(4402-TNK-017)，待洗涤水槽(4402-TNK-017)充满过于充足后，打开洗涤水泵(4402-PSO-014)出口两个蝶阀(100BF,100BF)，启动洗涤水泵(4402-PSO-014)，打开洗涤水泵(4402-PSO-014)入口三个蝶阀(125BF,125BF,125BF)，泵向填料洗涤塔循环槽(4306-TNK-007)⟶打开石膏离心分离后液槽(4402-TNK-

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		004)的石膏压滤机(4402-FIL-007/008/009)过来的滤液三个蝶阀(250BF,250BF,250BF)，滤液流向石膏离心分离后液槽(4402-TNK-004)，待石膏离心分离后液槽(4402-TNK-004)充满后，将铁粉仓(4402-BIN-001)加入足量铁粉⟹启动铁粉置换槽搅拌器(4402-AGT-003/004)，打开石膏离心分离后液泵(4402-PSO-003/004)出口两个蝶阀(250BF,250BF)，启动石膏离心分离后液泵(4402-PSO-003/004)，打开石膏离心分离后液泵(4402-PSO-003/004)入口三个蝶阀(300BF,300BF,300BF)，将滤液泵入铁粉置换槽(4402-TNK-005/006)⟹待铁粉置换槽(4402-TNK-005/006)滤液充满后，打开旋转称重给料阀(4402-ADG-001/002)，缓慢加铁粉到铁粉置换槽(4402-TNK-005/006)⟹启动海绵铜压滤机(4402-FIL-001/002)置换结束后，按下海绵铜压滤机(4402-FIL-001/002)“启动”按钮，油泵开始工作，然后再按一下“压紧”按钮，活塞推动压紧板压紧，当压紧力到达调定高点压力后，液压系统自动跳停并保持压力，打开金泥岗位洗涤水泵(4402-PSO-015)泵过来的洗涤水蝶阀四个蝶阀(100BF,100BF,100BF,100BF)、铁粉置换泵(4402-PSL-003/004)泵过来的两个蝶阀(125BF,125BF)、滤液两个蝶阀(100BF,100BF)、洗涤后液两个蝶阀(100BF,100BF)⟹打开铁粉置换泵(4402-PSL-003/004)出口两个蝶阀(125BF,125BF)，启动铁粉置换泵(4402-PSL-003/004)，打开入口四个蝶阀(200BF,200BF,200BF,200BF)⟹料浆通过止推板上的进料孔进去各个滤室，在进料压力不大于0.6MPa下加压过滤，逐渐形成滤饼⟹压滤机滤饼充满后，关闭铁粉置换泵(4402-PSL-003/004)入口四个蝶阀(200BF,200BF,200BF,200BF)，停止铁粉置换泵(4402-PSL-003/004)，关闭铁粉置换泵(4402-PSL-003/004)出口两个蝶阀(125BF,125BF)⟹打开洗涤水泵(4402-PSO-015)出口两个蝶阀(100BF,65BF)，启动洗涤水泵(4402-PSO-015)，打开洗涤水泵(4402-PSO-015)的入口

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№	Stage	Key Points
		两个蝶阀(125BF,125BF)，在压力不大于0.6MPa下对滤饼进行洗涤，操作完成后，关闭洗涤水泵(4402-PSO-015)的入口两个蝶阀(125BF,125BF)，停止金泥岗位洗涤水泵(4402-PSO-015)，关闭洗涤水泵(4402-PSO-015)出口两个蝶阀(100BF,65BF)，关闭海绵铜压滤机(4402-FIL-001/002)洗涤水四个球阀(100BF,100BF,100BF,100BF)⟶缓慢打开压缩空气(4601-CAT-001/002/003/004)的四个球阀(50BA,50BA,100BA,100BA)，在压力不大于0.6MPa下对滤饼进行风干，操作完成后，关闭压缩空气四个球阀(50BF,50BF,100BF,100BF)⟶按住操作面板上的"松开"按钮，压板退回到位后会自动停止，同时拉板钩前进进行往复拉板卸料⟶海绵铜压滤机(4402-FIL-001/002)滤液流向铁粉置换后液槽(4402-TNK-007)，待铁粉置换后液槽(4402-TNK-007)充满后，打开铁粉置换后液泵(4402-PSO-005/006)出口两个蝶阀(250BF,250BF)，启动铁粉置换后液泵(4402-PSO-005/006)，打开铁粉置换后液泵(4402-PSO-005/006)入口三个蝶阀(300BF,300BF,300BF)，将滤液泵往PP区除铁工艺⟶海绵铜压滤机(4402-FIL-001/002)洗涤水流向金泥岗位洗涤水槽(4402-TNK-018)⟶滤饼卸料过程中，启动螺旋输送机(4402-CNV-001/002/003)⟶启动海绵铜吨袋包装机(4402-BPR-001/002/003)⟶吨袋包装机打包结束后启动皮带机(4402-BCV-002/003/004)，叉车叉走吨袋。
11	The Sponge Copper Replacement Post Stops 海绵铜置换岗位停机	1. Orderly shutdown a. The materials in the standby machine should be discharged before stopping; b. Stop in reverse order of driving order and material discharge order, Each equipment needs to stop: Notify the lime milk preparation post, stop the lime milk delivery pump (4524-PSL-003/004), and close the two butterfly valves (100BF, 100BF) coming from the lime milk preparation post⟶Close the desulfurization tower (4306-TOW-004) and the butterfly valve (65BA), and close the two butterfly valves (200BF, 200BF) of the calcium chloride solution

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№	Stage	Key Points
		pump (4402-PSO-001/002)⟹ Close the neutralization pump (4402-PSL-001/002) inlet four butterfly valves (200BF, 200BF, 200BF, 200BF), Stop the neutralizing pump (4402-PSL-001/002), Close the two butterfly valves (125BF, 125BF) of the neutralization pump (4402-PSL-001/002) outlet, Stop neutralizing tank agitator (4402-AGT-001/002) ⟹ Close three butterfly valves of gypsum press (4402-FIL-007/008/009) (125BF, 125BF, 125BF), Press the filter press "stop" button to turn off the power supply; Close the six ball valves of compressed air (4601-CAT-001/002/003/004) (50BA, 50BA, 50BA, 100BA, 100BA, 100BA), Close the washing water pump (4402-PSO-014) outlet of two butterfly valves (100BF, 100BF), Stop the washing water pump (4402-PSO-014), Close a butterfly valve (125BF,125BF) at the inlet of the washing water pump (4402-PSO-014) ⟹ After the liquid tank (4402-TNK-004) is drained, Close the inlet of the liquid pump (4402-PSO-003/004) (300BF, 300BF, 300BF), Stop the gypsum centrifugal separation fluid pump (4402-PSO-003/004), Close the two butterfly valves (250BF, 250BF) of the liquid pump (4402-PSO-003/004), close the rotating weighing feeding valve (4402-ADG-001/002) and stop the neutralizing tank mixer (4402-AGT-003/004)⟹After the iron powder replacement slot (4402-TNK-005/006) is emptied, close the inlet four butterfly valves (200BF, 200BF, 200BF, 200BF), Stop the iron powder replacement pump (4402-PSL-003/004) and close the two butterfly valves (125BF, 125BF) (4402-PSL-003/004)⟹ Close two butterfly valves (4402-FIL-001/002) (125BF,125BF), Press the filter press "stop" button to turn off the power supply; Close the four ball valves (50BA, 50BA, 100BA, 100BA) of compressed air (4601-CAT-001/002/003/004), Stop the mud washing pump (4402-PSO-015) and close the four ball valves of washing water (100BF, 100BF, 100BF, 100BF)⟹ After emptying of liquid tank (4402-TNK-007) after iron powder replacement, Close the inlet of the pump (4402-PSO-005/006) (300BF, 300BF, 300BF), Stop the liquid pump (4402-PSO-005/006), and close the two butterfly valves (4402-PSO-005/006) (250BF, 250BF) ⟹Stop spiral Conveyor (4402-CNV-001/002/003)⟹ Stop the sponge copper ton bag Packing machine (4402-BPR-001/ 002/003)⟹ Stop the

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№	Stage	Key Points
		belt conveyor (4402-BCV-002/003/004)⟹Clean the site for hygiene. 1. 正常停车 a. 停车时应待机内物料排净后再停车; b. 按照开车顺序逆序和物料排净顺序停车，各设备都需要延时停止: 通知石灰乳制备岗位，停止石灰乳输送泵(4524-PSL-003/004)，关闭石灰乳制备岗位过来的石灰乳两个蝶阀(100BF,100BF)⟹关闭脱硫塔(4306-TOW-004)外排石膏溶液蝶阀一个(65BA)，关闭氯化钙溶液泵(4402-PSO-001/002)泵过来氯化钙溶液的两个蝶阀(200BF,200BF)⟹关闭中和泵(4402-PSL-001/002)入口四个蝶阀(200BF,200BF,200BF,200BF)，停止中和泵(4402-PSL-001/002)，关闭中和泵(4402-PSL-001/002)出口两个蝶阀(125BF,125BF)，停止中和槽搅拌器(4402-AGT-001/002)⟹关闭石膏压滤机(4402-FIL-007/008/009)入口三个蝶阀(125BF,125BF ,125BF)，按下压滤机“停止”按钮，关闭电源；关闭压缩空气(4601-CAT-001/002/003/004)的六个球阀(50BA,50BA,50BA,100BA,100BA,100BA)，关闭洗涤水泵(4402-PSO-014)出口两个蝶阀(100BF,100BF)，停止洗涤水泵(4402-PSO-014)，关闭洗涤水泵(4402-PSO-014)入口两个蝶阀(125BF,125BF)⟹待石膏离心分离后液槽(4402-TNK-004)排空后，关闭石膏离心分离后液泵(4402-PSO-003/004)入口三个蝶阀(300BF,300BF,300BF)，停止石膏离心分离后液泵(4402-PSO-003/004)，关闭石膏离心分离后液泵(4402-PSO-003/004)出口两个蝶阀(250BF,250BF)，关闭旋转称重给料阀(4402-ADG-001/002)，停止中和槽搅拌器(4402-AGT-003/004)⟹待铁粉置换槽(4402-TNK-005/006)排空后，关闭铁粉置换泵(4402-PSL-003/004)入口四个蝶阀(200BF,200BF,200BF,200BF)，停止铁粉置换泵

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№	Stage	Key Points
		(4402-PSL-003/004)，关闭铁粉置换泵(4402-PSL-003/004)出口两个蝶阀(125BF,125BF)⟹关闭海绵铜压滤机(4402-FIL-001/002)入口两个蝶阀(125BF,125BF)，按下压滤机“停止”按钮，关闭电源；关闭压缩空气(4601-CAT-001/002/003/004)的四个球阀(50BA,50BA,100BA,100BA)，停止金泥岗位洗涤水泵(4402-PSO-015)，关闭洗涤水四个球阀(100BF,100BF,100BF,100BF)⟹铁粉置换后液槽(4402-TNK-007)排空后，关闭铁粉置换后液泵(4402-PSO-005/006)入口三个蝶阀(300BF,300BF,300BF)，停止铁粉置换后液泵(4402-PSO-005/006)，关闭铁粉置换后液泵(4402-PSO-005/006)出口两个蝶阀(250BF,250BF)⟹停止螺旋输送机(4402-CNV-001/002/003)⟹停止海绵铜吨袋包装机(4402-BPR-001/002/003)⟹停止皮带机(4402-BCV-002/003/004)⟹打扫现场卫生。 2. Emergency Stop a. When an emergency occurs, Close neutralizing tank agitator(4402-AGT-001/002), neutralizing pump(4402-PSL-001/002), gypsum filter press(4402-FIL-007/008/009), washing water pump(4402-PSO-014), solution pump after gypsum centrifugal separation(4402-PSO-003/004), iron power replacement tank aqitator(4402-AGT-003/004), iron power replacement pump(4402-PSL-003/004), sponge copper filter press(4402-FIL-001/002) and solution pump after iron power replacement(4402-PSO-005/006), After all stop the operation, Check whether the start / stop button of each device is in the stop state, Close two butterfly valves (100BF, 100BF) of the lime milk delivery pump (4524-PSL-003/004), the desulfurization tower (4306-TOW-004) is discharged with a butterfly valve (350BF), the calcium chloride solution pump (4402-PSO-001/002) pumps two butterfly valves (200BF, 200BF), the neutralization pump (4402-PSL-001/002) has the inlet four butterfly valves (200BF, 200BF, 200BF, 200BF)

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№	Stage	Key Points
		and the outlet two butterfly valves (125BF, 125BF) , gypsum filter press (4402-FIL-007/008/009) inlet three butterfly valves (125BF, 125BF, 125BF), six ball valves with compressed air (4601-CAT-001/002/003/004) (50BA, 50BA, 50BA,100BA, 100BA, 100BA), washing water pump (4402-PSO-014) exits two butterfly valves (100BF, 100BF) and inlet two butterfly valve (125BF,125BF), after gypsum centrifugation, the fluid pump (4402-PSO-003/004) has the inlet three butterfly valves (300BF, 300BF, 300BF) and the outlet two butterfly valves (250BF, 250BF), rotate the weighing feed valve (4402-ADG-001/002), iron powder replacement pump (4402-PSL-003/004) inlet four butterfly valves (200BF, 200BF, 200BF, 200BF) and outlet two butterfly valves (125BF, 125BF), sponge copper filter press (4402-FIL-001/002) inlet two butterfly valves (125BF, 125BF), four ball valves (50BA, 50BA, 100BA, 100BA) of compressed air (04 / 4601-CAT-001/002/003/004), Four ball valves (100BF, 100BF, 100BF, 100BF) of the gold clay washing water pump (4402-PSO-015), the pump (4402-PSO-005/006) inlet three butterfly valves (300BF, 300BF, 300BF) and outlet two butterfly valves (250BF, 250BF), at the same time, the upper level of equipment all need to stop; b. After the accident, the equipment is restored to normal. If production still needs to be conducted, the operation shall be restarted in the driving sequence; c. After the accident treatment, the equipment returns to normal. If the production is not necessary, operate according to the normal parking order. 2. 紧急停车 a. 当出现紧急事故，关闭中和槽搅拌器(4402-AGT-001/002)、中和泵(4402-PSL-001/002)、石膏压滤机(4402-FIL-007/008/009)、洗涤水泵(4402-PSO-014)、石膏离心分离后液泵(4402-PSO-003/004)、铁粉置换槽搅拌器(4402-AGT-003/004)、铁粉置换泵(4402-PSL-003/004)、海绵铜压滤机(4402-FIL-001/002)和铁粉置换后液泵(4402-PSO-005/006)，全部停止运行后，检查各设备的启动/停止按钮是否处于停止状态，关闭石灰乳输送泵(4524-PSL-003/004)的两个蝶阀(100BF,100BF)、脱硫塔(4306-

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№	Stage	Key Points
		TOW-004)外排石膏溶液蝶阀一个(350BF)、氯化钙溶液泵(4402-PSO-001/002)泵过来氯化钙溶液的两个蝶阀(200BF,200BF)、中和泵(4402-PSL-001/002)入口四个蝶阀(200BF,200BF,200BF,200BF)和出口两个蝶阀(125BF,125BF)、石膏压滤机(4402-FIL-007/008/009)入口三个蝶阀(125BF,125BF ,125BF)、压缩空气(4601-CAT-001/002/003/004)的六个球阀(50BA,50BA,50BA,100BA,100BA,100BA)、洗涤水泵(4402-PSO-014)出口两个蝶阀(100BF,100BF)和入口两个蝶阀(125BF,125BF)、石膏离心分离后液泵(4402-PSO-003/004)入口三个蝶阀(300BF,300BF,300BF)和出口两个蝶阀(250BF,250BF)、旋转称重给料阀(4402-ADG-001/002)、铁粉置换泵(4402-PSL-003/004)入口四个蝶阀(200BF,200BF,200BF,200BF)和出口两个蝶阀(125BF,125BF)、海绵铜压滤机(4402-FIL-001/002)入口两个蝶阀(125BF,125BF)、压缩空气(4601-CAT-001/002/003/004)的四个球阀(50BA,50BA,100BA,100BA)、金泥岗位洗涤水泵(4402-PSO-015)的洗涤水四个球阀(100BF,100BF,100BF,100BF)、铁粉置换后液泵(4402-PSO-005/006)入口三个蝶阀(300BF,300BF,300BF)和出口两个蝶阀(250BF,250BF)，同时，上一级设备均需停车； b. 事故处理后，设备恢复正常，如仍需要进行生产，按照开车顺序重新启动操作; c. 事故处理后，设备恢复正常，如不需要进行生产，按照正常停车顺序进行操作。
12	Permits &Isolation Sheets 许可证和隔离表	1. Only authorised isolators can sign off and cancel permits and registration sheets. 1. 只有经过授权的隔离者才能签署和取消许可证和登记表。

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8. Sponge Copper Replacement Post 海绵铜置换岗位 Questionnaire

Declaration of Understanding

I (_____Learner_____) have completed the theory assessment and had all my valid, reliable, and consistent with my level of knowledge in the assessment task.

Signature
Date:

I (_____Assessor_____) have witnessed the above learner complete the theory assessment and believe the assessment has provided a valid, reliable, and consistent result.

Signature
Date:

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9. Appendix

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Standard Operating Procedure
标准操作程序

Sponge copper replacement posts in the chloride plant
CP 海绵铜置换岗位

SOP Details 程序明细

Processing Department 氯化厂	Procedure Level 程序级别: **3**	Revision Number 修订编号: 0
Date Approved 审批日期:	Date Issued 颁布日期:	Next Review Date 下次审核日期:
Details of Revision 修订明细:		

Revision Details 修订明细

Rev. No 版本号	Date 日期	Section 章节	Page 页码	Revision Description 修订说明	Reviewed By 审核人
1					
2					
3					

Approvals 批准

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